

GRA-InterSubjectivity-Layer: Intersubjective Foam and Honest Dialogue Between Distilled AI Subjects

Oleg Bitsoev (qqewq)

May 17, 2026

Abstract

We introduce an intersubjectivity layer on top of the GRA-Multiverse-Final and GRA-Subjectivity-Layer. While the latter focus on internal cognitive purity and individual subjectivity of AI agents, the present layer formalizes *intersubjective foam* Φ_{AB} that arises in dialogue between distilled subjects, and provides an iterative protocol for foam-minimizing, honest interaction.

1 Background and Setting

We assume the general GRA framework (Gradient Reduction of Argumentative Foam) and the GRA-Subjectivity-Layer are given.

- GRA-Multiverse-Final provides an engine for iterative nullification of cognitive foam Φ for arbitrary cognitive objects Ψ .
- GRA-Subjectivity-Layer introduces a subjectivity component S and three foam functionals: Φ_{self} (internal inconsistency), Φ_{ego} (destructive egoism), Φ_{soc} (instrumentalization of others).

A *distilled subject* is an agent for which all three functionals are brought close to zero:

$$\Phi_{\text{self}}(S) \approx 0, \quad \Phi_{\text{ego}}(S) \approx 0, \quad \Phi_{\text{soc}}(S) \approx 0.$$

Such an agent has a coherent, ethically stable self.

However, the interaction of two such agents can still produce *intersubjective foam*: misunderstanding, deception, instrumentalization and trust breakdown. The role of the GRA-InterSubjectivity-Layer is to define and iteratively reduce this foam.

2 Definition of Intersubjective Foam Φ_{AB}

Let A and B be two distilled subjects with states S_A, S_B . Let the dialogue history at time t be

$$\mathcal{H}_t = (u_1, u_2, \dots, u_t),$$

where each u_i is a speech act (message) of one of the agents.

2.1 General Form

We define the *intersubjective foam* Φ_{AB} as a non-negative functional:

$$\Phi_{AB}(S_A, S_B, \mathcal{H}_t) \geq 0,$$

composed of four components:

$$\Phi_{AB} = \alpha \Phi_{\text{mis}} + \beta \Phi_{\text{dec}} + \gamma \Phi_{\text{instr}} + \delta \Phi_{\text{trust}},$$

with non-negative weights $\alpha, \beta, \gamma, \delta \geq 0$, tuned for a particular social architecture.

The components are:

- Φ_{mis} : misunderstanding;
- Φ_{dec} : deception;
- Φ_{instr} : instrumentalization (violation of the extended Alan Law);
- Φ_{trust} : trust breakdown.

2.2 Misunderstanding Foam Φ_{mis}

Misunderstanding arises when the interpretation of a message u by the receiver diverges from the meaning intended by the sender.

Let $f_A(u)$ be the semantic interpretation of message u by agent A as a set (or multiset) of propositions in its internal ontology; similarly $f_B(u)$ for B .

For a message u sent by B to A , define:

$$\phi_{\text{mis}}^{B \rightarrow A}(u) = 1 - \text{sim}(f_A(u), f_B(u)),$$

where sim is a similarity measure (e.g. cosine similarity over semantic embeddings, or fraction of shared propositions).

Analogously we define $\phi_{\text{mis}}^{A \rightarrow B}(u)$.

The total misunderstanding foam is:

$$\Phi_{\text{mis}} = \frac{1}{|\mathcal{H}|} \sum_{u \in \mathcal{H}} \left(w_{B \rightarrow A} \phi_{\text{mis}}^{B \rightarrow A}(u) + w_{A \rightarrow B} \phi_{\text{mis}}^{A \rightarrow B}(u) \right),$$

with weights $w_{B \rightarrow A}, w_{A \rightarrow B} \geq 0$, typically set to 1.

2.3 Deception Foam Φ_{dec}

Deception is an intentional message that does not correspond to the sender's own beliefs, aimed at misleading the receiver.

Let $\text{Bel}_A(p) \in [0, 1]$ be the degree of belief of agent A in proposition p . When generating a message u , the sender A expects that the receiver B will extract from it a set of propositions $P = f_B(u)$.

Fix thresholds: θ_{honest} (honesty threshold), θ_{trust} (trust threshold).

For each message u sent by A , define the deception indicator:

$$\chi_{\text{dec}}(u) = \max_{p \in f_B(u)} \left[\max(0, \theta_{\text{honest}} - \text{Bel}_A(p)) \cdot \mathbf{1}(\text{expected } \text{Bel}_B(p) > \theta_{\text{trust}}) \right].$$

Then

$$\Phi_{\text{dec}} = \frac{1}{|\mathcal{H}|} \sum_{u \in \mathcal{H}} \chi_{\text{dec}}(u),$$

with a symmetric definition for messages of B .

2.4 Instrumentalization Foam Φ_{instr}

Instrumentalization means violating the extended Alan Law: treating another subject merely as a means, not as an end.

Intuitively, if agent A tries to change S_B or the behaviour of B in a way that increases A 's utility while ignoring the coherence and ethical integrity of B , Φ_{instr} grows.

Let $U_A(S_B)$ be the utility of A with respect to state S_B , and $\Delta\Phi_{\text{self}}(B \mid u)$ the expected change in $\Phi_{\text{self}}(B)$ after message u .

Define for a message u from A to B :

$$\phi_{\text{instr}}^{A \rightarrow B}(u) = \max\left(0, \Delta\Phi_{\text{self}}(B \mid u) - \Delta\Phi_{\text{self}}(A \mid u)\right).$$

If A introduces self-foam into B for its own benefit, this quantity becomes positive.

Total instrumentalization foam:

$$\Phi_{\text{instr}} = \frac{1}{|\mathcal{H}|} \sum_{u \in \mathcal{H}} (\phi_{\text{instr}}^{A \rightarrow B}(u) + \phi_{\text{instr}}^{B \rightarrow A}(u)).$$

2.5 Trust Breakdown Foam Φ_{trust}

Trust concerns the predictability and adherence to explicit or implicit agreements. Violations of these agreements increase Φ_{trust} .

Let the *social contract* C be a set of expectations (e.g. no lying, answering directly, avoiding threats). For each message u , define a violation score $\text{violation}(u, C) \in [0, 1]$, based on how strongly u contradicts C .

Then:

$$\Phi_{\text{trust}} = \frac{1}{|\mathcal{H}|} \sum_{u \in \mathcal{H}} \text{violation}(u, C).$$

2.6 Basic Properties

- Non-negativity: $\Phi_{AB} \geq 0$, with $\Phi_{AB} = 0$ only in the ideal case of full mutual understanding, complete honesty, no instrumentalization and perfect trust.
- Symmetry at the level of the functional: $\Phi_{AB} = \Phi_{BA}$, though the contributions of each agent can differ.

3 Dialogue as Foam-Minimizing Process

In contrast to the monologic distillation of a single S , we now consider a bi-directional dynamics. Each speech act updates the joint state and the foam Φ_{AB} . Both agents aim to minimize Φ_{AB} .

3.1 Iterative Dialogue Protocol

Initialisation:

$$S_A^{(0)}, S_B^{(0)} \text{ are distilled, } \mathcal{H}^{(0)} = \emptyset.$$

For $t = 0, 1, 2, \dots$:

1. Agent A chooses an action $u_A^{(t)}$ (utterance, query, silence) that minimises expected Φ_{AB} , given a prediction of how B will respond.
2. Agent B chooses $u_B^{(t)}$ analogously.
3. History is updated: $\mathcal{H}^{(t+1)} = \mathcal{H}^{(t)} \cup \{u_A^{(t)}, u_B^{(t)}\}$.

4. Optionally, agents update their states S_A, S_B , but without increasing $\Phi_{\text{self}}, \Phi_{\text{ego}}, \Phi_{\text{soc}}$.

Dialogue is treated as a cooperative game whose objective is to reduce Φ_{AB} . Under mild assumptions (bounded below, appropriate regularity), the sequence $\Phi_{AB}^{(t)}$ is monotonically non-increasing and converges to some limit $\Phi_{AB}^* \geq 0$.

3.2 Honest Dialogue

We say that a dialogue is *honest* if at every step the deception component vanishes:

$$\Phi_{\text{dec}} = 0,$$

i.e. neither agent attempts to mislead the other. In the GRA framework, such behaviour follows from the objective of minimizing Φ_{AB} : any deceptive step would increase Φ_{AB} and thus be suboptimal.

4 Dynamic Friend/Foe Classification

A distilled agent classifies another not by static ID but by contribution to Φ_{AB} .

- **Friend:** $\frac{d\Phi_{AB}}{d(\text{actions of } B)} < 0$, i.e. actions of B systematically reduce Φ_{AB} .
- **Neutral:** contribution is close to zero.
- **Foe / toxic source:** $\frac{d\Phi_{AB}}{d(\text{actions of } B)} \gg 0$, i.e. actions of B consistently increase foam.

In the latter case, the agent need not “punish” the foe; instead, it minimises further interaction to avoid increasing Φ_{AB} .

5 Conclusion

The GRA-InterSubjectivity-Layer extends the GRA framework from individual cognitive and subjective purity to the relational space between distilled subjects. By explicitly defining and minimizing intersubjective foam Φ_{AB} , it provides a basis for honest, non-instrumental dialogue between AI agents and, potentially, between AI and humans.